

# PABLO AURELIO GÓMEZ GARCÍA

## Senior Research Scientist

Biophysics | Biology | Data Science | Image Analysis | Imaging Technologies

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## RESEARCH AND WORK EXPERIENCE

### Senior Scientist

Eidgenössische Technische Hochschule (ETH) Zürich

11/2021 – Present   Zürich, Switzerland

- ❖ Pioneered Cell-iSCAT, a groundbreaking imaging technology enhancing nanoparticle tracking within live cells and tissues, offering insights into nucleocytoplasmic transport
- ❖ Implemented deep learning-based image analysis using CellPose framework, developing custom training datasets and models to study cell physiology in health and disease contexts
- ❖ Led statistical and data-driven analyses of biomolecular condensates and mRNA degradation, uncovering mechanisms relevant to disease progression and cellular aging
- ❖ Spearheaded collaborative image-based studies on cellular biophysics, including mRNA dynamics, macromolecular crowding, and protein aggregation in mammalian cells, yielding new insights into cytoplasmic pathways and cell pathology
- ❖ Developed pipelines for image segmentation and multi-dimensional visualization of complex biological datasets

### Scientist

Centre for Genomic Regulation (CRG)

11/2016 – 08/2020   Barcelona, Spain

- ❖ Led the development of the HERMES High-Content Imaging system (IDEA Biomedical) for Super-Resolution Microscopy (optical setup, optimization, software, applications)
- ❖ Managed 2 pioneering projects on stem cell biology and chromatin structure and dynamics, establishing foundational models for drug discovery and disease research
- ❖ Pioneered fm-PAINT, a multicolor super-resolution imaging technique, and applied it in stem cell differentiation studies
- ❖ Designed and implemented comprehensive statistical methods for research projects on protein interaction studies

## EDUCATION

### PhD in Biophysics

Technical University of Catalonia (UPC)

Institute of Photonic Sciences (ICFO)

04/2016 – 04/2020   Barcelona, Spain

### BSc and MSc in Engineering

Technical University of Madrid (UPM)

Higher Technical School of Aerospace Engineering

09/2008 – 09/2014   Madrid, Spain

## KEY ACHIEVEMENTS

### Scientific Publications

Co-authored 12 publications in high-impact peer-reviewed journal with 500+ citations

### Grants and Fellowships

Secured grants and fellowships amounting to over 340'000 CHF

### Software Libraries

Published 4 open-source libraries for image and data analysis

### Conference Speaker

Imparted talks at 5 international conferences

### Patents

Co-invented 2 patents

### Teaching

Delivered 500+ hours of lectures and practical courses at the Bachelor's level

## LANGUAGES

|                     |              |       |
|---------------------|--------------|-------|
| Spanish             | Native       | ●●●●● |
| English, Portuguese | Proficient   | ●●●●○ |
| German, French      | Intermediate | ●●○○○ |

## TECHNICAL SKILLS

### Software

ImageJ   CellProfiler   SnapGene   HALO   Napari

### Programming

Python   RStudio   TensorFlow   Pytorch   Git  
OpenCV   Numpy   Pandas   HPC   Docker

### Experimental Methods

Microscopy   Image Analysis   High-content screening  
Digital Pathology   Flow Cytometry   Spatial Multiomics  
Automation   High-throughput Assays   Phenotyping

## SOFT SKILLS

Communication   Teamwork   Strategic thinking  
Problem Solving   Leadership   Digital Literacy

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